Summary

This project uses the Formula 1 World Championship dataset from 1950-2024, which includes a lot of information about how the sport has developed over time. The dataset includes tables for drivers, constructors (basically teams), races, and seasons in which the races took place. It also includes more detailed data like race results, qualifying results, lap times, pit stops, and both driver and constructor standings. Overall, this dataset gives a clear view of how a Formula 1 race operates, starting from individual races and going all the way up to the final results of the season.

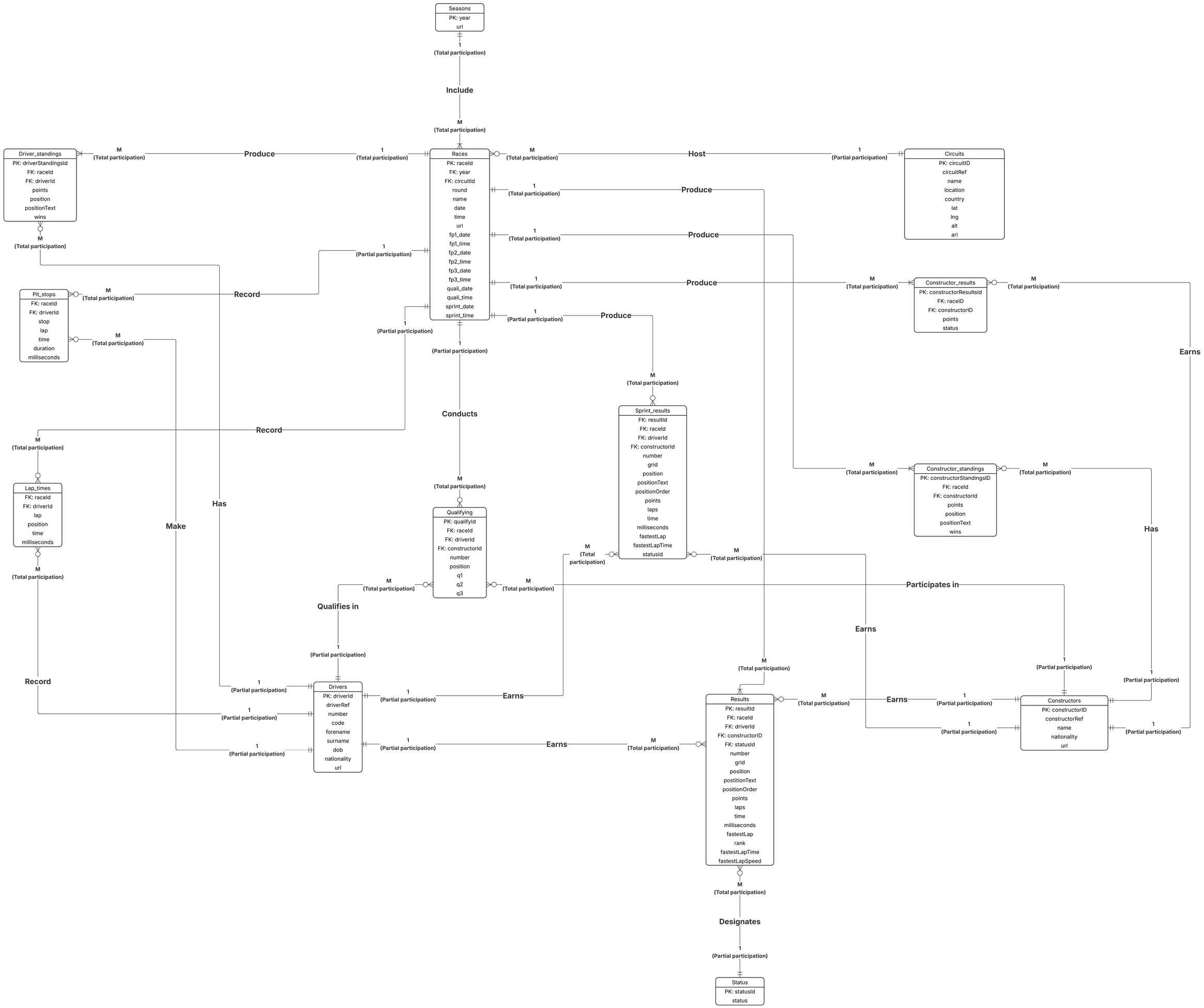
The reason for choosing this dataset is firstly because some of our team members adore and are huge fans of Formula 1, so they were already somewhat familiar with it which helped the team acquire adequate knowledge on how this sport works. The dataset itself has many natural relationships between different parts of the data. For instance, a driver competes in many races, represents a constructor, and earns results and points across a season. Races take place at specific circuits and belong to one specific season. All of these easy-to-comprehend connections make it a strong dataset for building and ER diagram, since it is significantly straight forward to determine which entities have a many-to-one, one-to-one, or many-to-many relationship. It is also more interesting to work with real world data rather than fictional ones.

The dataset contains 14 tables and over 700,000 rows, so it is large enough to support complex queries and deeper analysis. At the same time, it is also quite tractable to work with since the tables are connected using common fields such as raceId, driverId, constructorId, and circuitId, which keeps everything packed and linked together. Most of the tables are connected to more than one other table, which helps avoid a simple “one main table” structure and instead creates a more complete and connected database design. This dataset also is not a wheel-and-spoke model, as the data is distributed across multiple interconnected tables rather than centred around a single one. Some tables store really important and core information while others store detailed event data like lap times, pit stops, and results.

Moreover, another reason that we opted for this dataset is that it is fairly clean and structured. Most of the data is separated into logical and meaningful tables, which helps when converting it into a relational model and normalizing it i.e. the driver’s information is stored separately from race results, and race information is stored separately from circuits. This reduces data redundancy and makes the database easier to organize and understand.

Dataset taken from Kaggle:

https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020?resource=download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FK: circuitId
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time
url
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fp1_time
fp2_date
fp2_time
fp3_date
fp3_time
quali_date
quali_time
sprint_date
sprint_time
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Circuits
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circuitRef
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location
country
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lng
alt
arl
Pit_stops
FK: raceId
FK: driverId
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lap
time
duration
milliseconds
Record
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Earns
Sprint_results
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positionText
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positionText
wins
Has
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time
milliseconds
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FK: driverId
FK: constructorId
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position
q1
q2
q3
Participates in
Qualifies in
Drivers
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driverRef
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code
forename
surname
dob
nationality
url
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grid
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positionOrder
points
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millisecondes
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rank
fastestLapTime
fastestLapSpeed
Constructors
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constructorRef
name
nationality
url
Designates
Status
PK: statusId
status
M
(Total participation)
1
(Total participation)
1
(Partial participation)
M
(Partial participation)

ERD "Crow's Foot" Relationship Symbols [Quick Reference]

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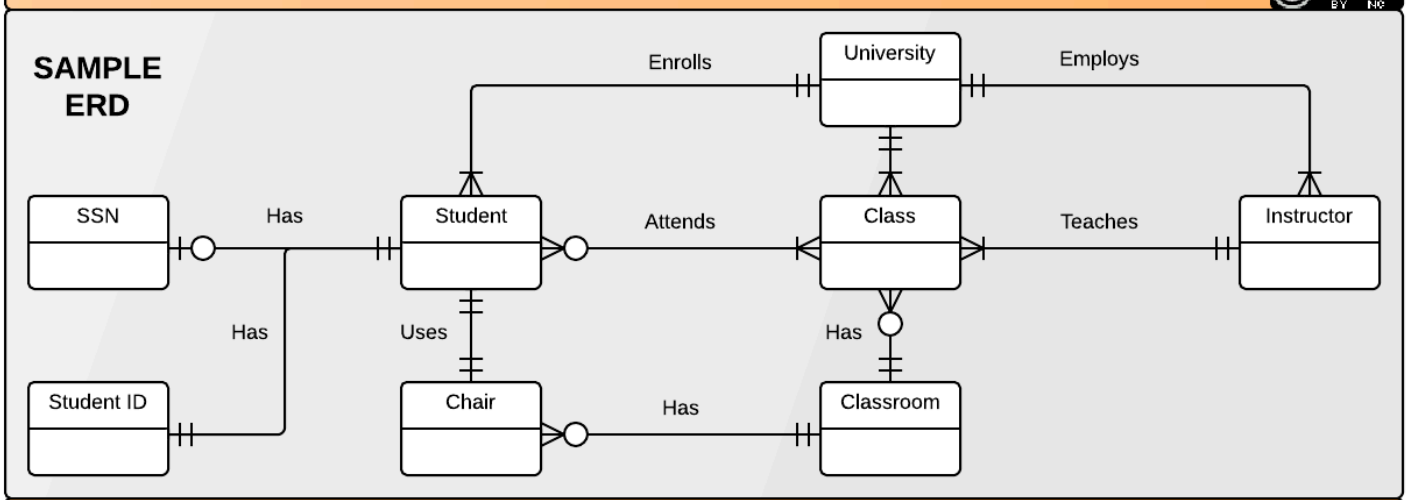

Notation	Meaning	Example
	Relationship	Student — Enrolls — University
	One	Student — Has — Student ID Number
	Many	Student — Attends — Class
	One and ONLY One	Student — Uses — Chair
	Zero or One	Student — Has — Social Security Number
	One or Many	Instructor — Teaches — Class
	Zero or Many	Classroom — Has — Chair

Seasons(Year, URL)

PK Year

Circuits(CircuitID, CircuitRef, Name, Location, Country, Lat, Lng, Alt, URL)

PK CircuitID

Drivers(DriverID, DriverRef, Number, Code, Forename, Surname, DOB, Nationality, URL)

PK DriverID

Constructors(ConstructorID, ConstructorRef, Name, Nationality, URL)

PK ConstructorID

Status(StatusID, Status)

PK StatusID

Races(RaceID, Year, CircuitID, Round, Name, Date, Time, URL, FP1_Date, FP1_Time, FP2_Date, FP2_Time, FP3_Date, FP3_Time, Quali_Date, Quali_Time, Sprint_Date, Sprint_Time)

PK RaceID

FK Year REF Seasons(Year)

FK CircuitID REF Circuits(CircuitID)

Results(ResultID, RaceID, DriverID, ConstructorID, StatusID, Number, Grid, Position, PositionText, PositionOrder, Points, Laps, Time, Milliseconds, FastestLap, Rank, FastestLapTime, FastestLapSpeed)

PK ResultID

FK RaceID REF Races(RaceID)

FK DriverID REF Drivers(DriverID)

FK ConstructorID REF Constructors(ConstructorID)

FK StatusID REF Status(StatusID)

Qualifying(QualifyID, RaceID, DriverID, ConstructorID, Number, Position, Q1, Q2, Q3)

PK QualifyID

FK RaceID REF Races(RaceID)

FK DriverID REF Drivers(DriverID)

FK ConstructorID REF Constructors(ConstructorID)

Driver_Standings(DriverStandingsID, RaceID, DriverID, Points, Position, PositionText, Wins)

PK DriverStandingsID

FK RaceID REF Races(RaceID)

FK DriverID REF Drivers(DriverID)

Constructor_Standings(ConstructorStandingsID, RaceID, ConstructorID, Points, Position, PositionText, Wins)

PK ConstructorStandingsID

FK RaceID REF Races(RaceID)

FK ConstructorID REF Constructors(ConstructorID)

Constructor_Results(ConstructorResultsID, RaceID, ConstructorID, Points, Status)

PK ConstructorResultsID

FK RaceID REF Races(RaceID)

FK ConstructorID REF Constructors(ConstructorID)

Lap_Times(RaceID, DriverID, Lap, Position, Time, Milliseconds)

PK (RaceID, DriverID, Lap)

FK RaceID REF Races(RaceID)

FK DriverID REF Drivers(DriverID)

Pit_Stops(<u>RaceID, DriverID, Stop</u>, Lap, Time, Duration, Milliseconds)

PK (RaceID, DriverID, Stop)

FK RaceID REF Races(RaceID)

FK DriverID REF Drivers(DriverID)

Sprint_Results(<u>ResultID</u>, RaceID, DriverID, ConstructorID, StatusID, Number, Grid, Position, PositionText, PositionOrder, Points, Laps, Time, Milliseconds, FastestLap, FastestLapTime)

PK ResultID

FK RaceID REF Races(RaceID)

FK DriverID REF Drivers(DriverID)

FK ConstructorID REF Constructors(ConstructorID)

FK StatusID REF Status(StatusID)

*since ResultID is unique for each row so I let ResultID is PK

Cardinality and participation justifications

Seasons - Races

A season includes many races, and each race belongs to exactly one season. Race has total participation because every race must occur in a season. Season has total in this dataset because each stored season has races

Circuits - Races

A circuit may host zero or many races, while each race must be held at exactly one circuit. Therefore, the relationship is one-to-many, with partial participation on the circuit side and total participation on the race side.

Races - Results

A race produces many results, and each result belongs to exactly one race. Each result must be associated with a race, and every recorded race has at least one result, so participation is total on both sides in the dataset.

Drivers - Results

A driver may earn zero or many results across different races, while each result belongs to exactly one driver. Result has total participation, while driver has partial participation because a driver may exist in the dataset without appearing in every race.

Constructors - Results

A constructor may earn zero or many results, while each result belongs to exactly one constructor. Result has total participation, and constructor has partial participation.

Status - Results

A status may describe many results, while each result has exactly one status. Result has total participation because every result must have a status, while status has partial participation because some status values may not appear in a given subset of the dataset.

Races - Driver_standings

A race produces many driver standings, and each driver standing belongs to exactly one race. Driver standing has total participation. Race can be justified as total in this dataset because driver standings are recorded after each race.

Drivers - Driver_standings

A driver may have zero or many driver standings over time, while each driver standing belongs to exactly one driver. Driver standing has total participation, while driver has partial participation because a driver may have no standing records yet.

Races - Constructor_standings

A race produces many constructor standings, and each constructor standing belongs to exactly one race. Constructor standing has total participation. Race is treated as total in this dataset because constructor standings are recorded after each race.

Constructors - Constructor_standings

A constructor may have zero or many constructor standings, while each constructor standing belongs to exactly one constructor. Constructor standing has total participation, while constructor has partial participation.

Races - Constructor_results

A race produces many constructor results, and each constructor result belongs to exactly one race. Constructor result has total participation, and race has total participation in the dataset because constructor-level race results are recorded for each race.

Constructors - Constructor_results

A constructor may earn zero or many constructor results, while each constructor result belongs to exactly one constructor. Constructor result has total participation, while constructor has partial participation.

Races - Qualifying

A race conducts zero or many qualifying sessions, and qualifying records belong to exactly one race. Qualifying has total participation. Race may be treated as partial because not every race in the dataset must have qualifying data stored

Drivers - Qualifying

A driver may qualify in zero or many races, while each qualifying record belongs to exactly one driver. Qualifying has total participation, while driver has partial participation.

Constructors - Qualifying

A constructor may participate in zero or many qualifying sessions, while each qualifying record belongs to exactly one constructor. Qualifying has total participation, while constructor has partial participation.

Races - Lap_times

A race records zero or many lap times, and each lap time belongs to exactly one race. Lap time has total participation. Race is partial because some races may not have lap-time data recorded in all context.

Drivers - Lap_times

A driver may record zero or many lap times, while each lap time belongs to exactly one driver. Lap time has total participation, while driver has partial participation.

Races - Pit_stops

A race records many pit stops, and each pit stop belongs to exactly one race. Pit stop has total participation. Race is partial because some races may have no recorded pit stops.

Drivers - Pit_stops

A driver may make zero or many pit stops, while each pit stop belongs to exactly one driver. Pit stop has total participation, while driver has partial participation.

Races - Sprint_results

A race may produce zero or many sprint results, while each sprint result belongs to exactly one race. Sprint result has total participation, while race has partial participation because not every race weekend includes a sprint event.

Drivers - Sprint_results

A driver may earn zero or many sprint results, while each sprint result belongs to exactly one driver. Sprint result has total participation, while driver has partial participation.

Constructors - Sprint_results

A constructor may earn zero or many sprint results, while each sprint result belongs to exactly one constructor. Sprint result has total participation, while constructor has partial participation.

References

- Dataset: Formula 1 World Championship (1950 - 2024)
- Information about Crow's foot ER symbols: Crow foot notation - Relationship Symbols and How to Read Diagrams
- Crow's foot ER notation legend: Interpreting the Crow's foot ER symbols